

NEUROLINK WEAR · SMART BAND HEALTH & SAFETY PLATFORM

Predictive health intelligence, worn every day.

One sensing band, one explainable AI engine and one caregiver dashboard — built to catch a health change between check-ins and turn it into a human response.

Real-time vitals · eight monitored conditions · per-wearer calibration · fall detection with GPS · emergency dispatch · caregiver back-office

LIVE WEARER SIGNAL

Margaret T. · 78 · at home

● LIVE

Heart rate

72 bpm

Blood oxygen

97 %

Skin temp

36.4 °C

Stress index

0.28 /1

8 conditions monitored · 2-tier safety rules · 24 h inactivity watch

PROPOSITION

Device + AI + care workflow

FIRST MARKET

Aging at home

STAGE

Working software prototype

ASK

\$1.5M pre-seed · pilot-led

Nine questions this deck answers

Each section is built to be read on its own — every figure carries its source and status.

01

Problem

The monitoring gap between check-ins

02

Market

Who pays, and why now

03

Solution

Sense → understand → respond

04

Product

Band, dashboard, safety workflow

05

Intelligence

Equations, calibration, LLM layer

06

Audience

Segments, personas, beachhead

07

Roadmap

Shipped, next, future features

08

Business

Model, pricing, go-to-market

09

Financials

5-year plan, break-even, ask

Status legend used throughout: ✓ shipped in the current build · ◦ roadmap candidate (not committed)

What exists, what is next, what it is worth

WORKING SOFTWARE PROTOTYPE — QA + CALIBRATION SUITES GREEN (141 · 74 · 17 CHECKS)

Real-time monitoring

Heart rate, blood oxygen, skin temperature, GSR stress index and IMU motion streamed live — 2-second WebSocket updates, 8 conditions evaluated on every reading.

Safety floors that never move

Two-tier rules: clinical absolutes OR the wearer's own σ -deviation. Personalisation can only add sensitivity.

Caregiver back-office built in

Contacts, device pairing with a staged MQTT connection test, thresholds, calibration wizard, roles and 30-day sessions.

STATUS HONESTY — WHAT IS NOT PROVEN YET

Wearable is prototype-grade hardware; alert precision, battery life, adherence and regulatory scope are hypotheses that only a field pilot can settle.

Explainable AI, no training set

Physiology equations plus per-wearer calibration replace a black box; every alert cites the numbers that triggered it.

End-to-end escalation

Incident timeline, live GPS map, dispatch log and one-click emergency-contact delivery in one safety view.

Measured, not asserted

141-check QA battery, 74-check calibration suite, 17 DOM checks — all passing, all re-runnable on demand.

THE ASK

\$1.5M pre-seed

~18 months to a validated, pilot-ready product — 2-3 pilot partners, 100-150 wearers, measured precision and response time.

01

PROBLEM & MARKET

The gap between check-ins

Continuous context and a response path — the problem worth solving first.

Signals without context

Trends that start quietly

Responses that are improvised

Critical health changes happen between check-ins

Care is delivered in snapshots. A short visit, a phone call, a reading taken once — while the signals that matter are continuous.

1

Signals lose their context

A heart rate of 96 bpm is unremarkable on a walk and alarming at rest. Single readings, taken alone, cannot separate the two.

2

Early warning is easy to miss

Fever, fatigue, panic and low oxygen usually begin as subtle trends — a slow drift that a person feels only after the fact.

3

The response path is improvised

When something does go wrong at 3 a.m., the people who care are called in an order nobody wrote down.

THE INSIGHT THAT SHAPES THE PRODUCT

The missing piece is not another measurement. It is context, timing and a response that already exists.

A DAY IN THE CURRENT MODEL

- 07:30 **Morning check-in call**
"I'm fine." — no data
- 13:10 **Blood pressure at the clinic**
one reading, archived
- 18:40 **A slow walk, breathlessness**
not noticed by anyone
- 02:15 **Night-time bathroom fall**
no alert, no context, no log

WHAT FAMILIES ACTUALLY NEED

- **A trend, not a number.** "Her resting pulse has been 12 bpm higher for two days."
- **A threshold built for one body.** Her normal is not the population average.
- **A path to a human.** When it matters, someone is reached — and it is logged.

Who pays, and why now

Three segments, one sensing platform — the first commercial wedge stays narrow on purpose.

BEACHHEAD

Aging-at-home households

Family caregiver households with one wearer and 1-3 trusted contacts. Direct subscription paid by a son or daughter.

- ~1 in 3 older adults wants to stay home
- Informal family care is the default safety net

EXPANSION

Home care & senior living

Multi-wearer dashboards for professional care teams, paid pilots, staff workflow and reporting.

- Staff-to-resident alerting and audit trail
- Buyer already budgets for monitoring

LONG TERM

Remote care ecosystem

Provider, payer and employer relationships once detection quality and integration are proven.

- Chronic-condition remote monitoring
- Escalation evidence unlocks reimbursement

WHY NOW — FIVE ENABLERS

Sensor economics — PPG, GSR, IMU and temperature modules are commodity parts.

Transparent AI — Equation-based models pass comprehension review where black boxes stall.

Open connectivity — MQTT over Wi-Fi plus a web dashboard removes the app-store gate.

Edge compute — An ESP32-class chip runs feature extraction and thresholds on the wrist.

Remote-care demand — Care-at-a-distance is now a mainstream family arrangement.

SCALE ILLUSTRATION — NOT A TAM CLAIM

100,000

paid wearer accounts

\$14.4M

annual subscription revenue at \$12/month

\$14.9M

one-off device revenue at \$149 ASP

Illustrative capacity case to show model mechanics only. Launch geography, eligible base, reimbursement and pricing require validation before any market-size statement.

The timing argument in one page

The demographic curve is not a forecast

It is arithmetic. Aging populations and caregiving shortages are policy problems already being funded.

Consumer wearables normalised the wrist

The behaviour change — charging a device, sleeping with it, reading its app — has already happened.

But consumer devices stop at data

Rings and watches report. They do not run a caregiver workflow, escalate to a contact or log a dispatch.

Clinical-grade transparency is now a requirement

Care organisations ask “why did it alert?” before they ask “how accurate is it?” Explainability is a sales asset.

THE GAP WE OCCUPY

Fitness trackers Data, no care workflow

Medical alert buttons Reactive, one-way, no context

Clinical monitoring Effective but institution-bound

NeuroLink Wear Continuous sensing + explanation + escalation

MARKET EVIDENCE TO COLLECT IN THE PILOT

- Willingness to pay: family subscription vs. device-only.
- Attach rate inside a home-care operator's caseload.
- Cost to serve per wearer per month (support dominates).
- Reimbursement pathway questions for the remote-care segment.
- Churn after the first false alert.

02

SOLUTION & PRODUCT

Device, intelligence, action

One sensing platform: a wearable band, an explainable AI engine and a caregiver workflow.

Five sensors on an ESP32

Two-tier detection

Six caregiver views shipped

Sense, understand, respond — then learn the wearer

A device, an intelligence layer and a care workflow that share one data spine.

1 SENSE

PPG, GSR, motion, temperature and GPS

The band samples heart rate, blood oxygen, HRV, skin conductance, skin temperature and 6-axis motion — plus location context — and publishes every 1-5 s over MQTT.

25 Hz motion sampling · 1 Hz vitals · published every 1-5 s

2 UNDERSTAND

Equations + calibration + language

Raw signals become features, features become equations, equations become a condition — with a plain-language explanation of the evidence behind it.

Evaluated on every reading at a ~2-second cadence

3 RESPOND

Explain, recommend, escalate

The caregiver sees what changed, why it matters and what to do — and can reach a human with one action, with the whole event logged.

notify → confirm → escalate → log, with GPS attached

THE LOOP CLOSES ON THE WEARER

Every reading and every acknowledged alert feeds back into the wearer's personal baselines, so the next threshold is better than the last.

Reading

>

Explained alert

>

Human response

>

Baseline update

>

Sharper alert

The last step is the business model: it gets better the longer it is worn.

WHAT MAKES IT DIFFERENT

- **No dataset dependency. Works on day one for a new wearer.**
- **Same math on device and server. The ESP32 can run the identical equations.**
- **Explained, not scored. Alerts name the signals and the deviation.**

Six views, one caregiver workflow

Shipped in the current build — a responsive, role-aware web app with live WebSocket data.



THE SIX SHIPPED VIEWS

Live Overview

Vitals, IMU motion, live feed, device health, AI insight, quick actions.

Alerts

Severity, evidence readings, explanation, recommendations, acknowledge / resolve.

Safety

Incident timeline, GPS map, inactivity watch, one-click dispatch with delivery log.

Trends

HRV, temperature & SpO₂, stress vs HR — 24 h / 48 h / 7 d / 30 d filters.

Management

Contacts, devices & MQTT, thresholds, calibration, wearer profile, users.

Settings

Broker configuration, simulator controls, theme, session and role handling.

Hardware: a five-sensor band on an ESP32

Components referenced by the current firmware source. Measurements are prototype-grade and require bench validation.

Component	Measures	Role in the system	Interface
MAX30102 — PPG	Heart rate, SpO ₂ , HRV	Cardiovascular context, oxygen and recovery signals	I ² C
GSR electrodes	Skin conductance (μS)	Sympathetic arousal; tonic / phasic EDA split	Analog
MPU6050 — IMU	Accel + gyro, 6 axes	Activity state, tremor, steps, fall signature	I ² C
MLX90614 — IR	Skin temperature	Fever trend and thermal drift, core-temperature estimate	I ² C (SMBus)
NEO-6M — GNSS	Latitude / longitude	Location context attached to every escalated event	UART
ESP32-WROOM	Edge compute + radio	Feature extraction, threshold filtering, MQTT publish	Wi-Fi / BLE

FIRMWARE BEHAVIOUR (CURRENT SOURCE)

- Publishes JSON over MQTT to `neurolink/wear/<id>/telemetry` every 1-5 seconds.
- Runs the same equations as the server — stress index, core-temperature estimate, fall physics.
- Local threshold filter so a broken link cannot silence a critical event.
- Two OLED surfaces for wearer-facing status and battery / link state.

PROTOTYPE → PRODUCT GATES

Accuracy — PPG and temperature validated against reference instruments

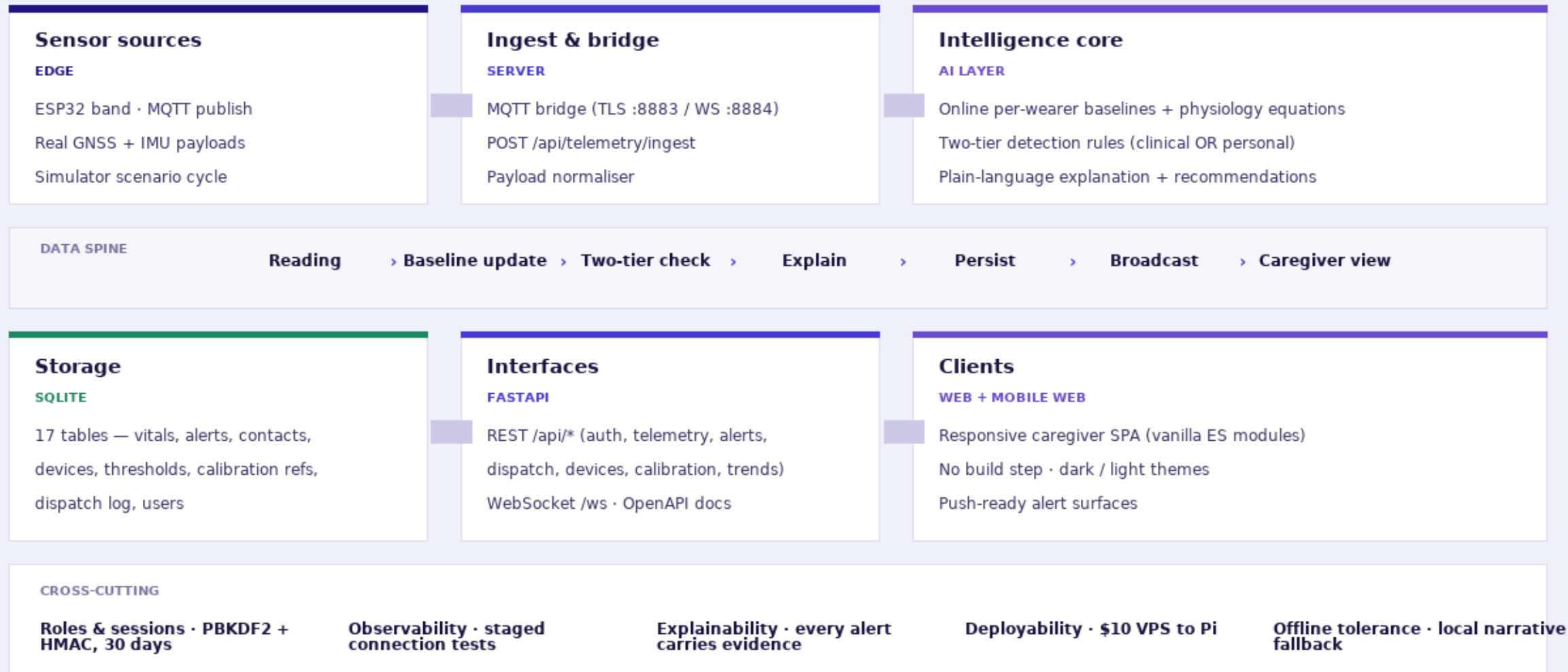
Battery — target multi-day wear between charges, measured per user

Form factor — wearable for 24 h including sleep, with a replaceable strap

Manufacturing — DFM review, component second-sourcing, EMC pre-scan

One ingest path for simulated and real devices

The end-to-end software path is implemented and running; device timing is the field-readiness gate.



03

INTELLIGENCE

Explainable intelligence

Equations and per-wearer calibration instead of an opaque model.

No training dataset required

Clinical floors that never move

Every alert cites its evidence

Explainable AI without a training dataset

Population ML needs data the wearer never generated. Physiological equations plus per-wearer calibration work on day one — and every number can be defended.

Equation	Form	What it answers
Robust z-score	$z = (x - \text{median}) / (1.4826 \cdot \text{MAD})$	How far is this reading from HER normal, resistant to outliers?
EDA split	$\text{tonic} = \text{EWMA}(\text{GSR}) \cdot \text{phasic} = \max(0, \text{GSR} - \text{tonic})$	Is this slow arousal or a sudden sympathetic spike?
Stress index	$S = \text{clamp}(0.5 + \Sigma \sigma\text{-evidence} / 5)$	Combined stress evidence from GSR, HRV and heart rate at rest
Core temperature	$T_{\text{core}} \approx \text{gain} \cdot T_{\text{skin}} + \text{offset}$	Skin reading converted to a core-equivalent, personally offset
Fever score	$\text{core level} + ^\circ\text{C/h slope} + \text{HR} \leftrightarrow \text{T coupling}$	Is this a real fever trend or a warm room?
Hypoxic burden	$B \leftarrow 0.85 \cdot B + \text{minutes} \cdot \max(0, 92 - \text{SpO}_2)$	Sustained desaturation instead of a single noisy sample
Fall physics	$\text{free-fall} \rightarrow \text{impact} \geq 2.8 \text{ g} \rightarrow \text{tumble} \geq 2.4 \text{ rad/s} \rightarrow \text{stillness}$	Four-stage signature that separates a fall from a bump
Expected HR	$\text{HR}_{\text{exp}} = \text{HR}_{\text{rest}} + \text{activity demand} \cdot \text{age factor}$	Is this pulse appropriate for what she is doing right now?

WHY IT MATTERS COMMERCIALY

- A new wearer is covered on day one — no cold-start gap.
- Every alert can be explained in a sentence to a clinician or a regulator.
- The same formulas run on the wrist, so cost does not scale with users.

LEGACY ML ASSETS (KEPT, SUPERSEDED)

An offline pipeline with Isolation Forest, an LSTM autoencoder and an activity classifier remains in the repository for research comparison.

1,875 training rows • 14 channels • IF threshold -0.02 • LSTM AE 0.332. Not used for live detection.

The model learns one body, not a population

Four sources of truth, tracked per baseline, with confidence that grows as the band is worn.

AUTO

Every stream tick

Slow EWMA baselines with a quiet gate, so a crisis reading never drags the baseline; robust MAD tracks personal noise.

AUTO-FIT

Last 24 h of data

Snapshots baselines to robust percentiles of stored vitals and shows a before / after diff before committing.

GUIDED

Reference instrument

Oral thermometer, clinical pulse oximeter, measured resting HR/HRV — each logged with a timestamp.

MANUAL

Clinician or caregiver

Direct overrides when a professional knows the wearer better than the data yet does.

WORKED EXAMPLE — GUIDED CORE-TEMPERATURE OFFSET

- 1 Nurse takes an oral temperature **37.1 °C**
- 2 Band reports skin temperature at that moment **36.5 °C**
- 3 Personal skin → core offset is stored **+0.6 °C**
- 4 Every later reading uses the personal offset **core-equiv. T**

Fever now triggers on a core-equivalent temperature, not on a warm forearm.

BASELINES TRACKED PER WEARER

Baseline	Example value	Source
Resting heart rate	68 bpm	auto
Resting HRV (RMSSD)	44 ms	guided
Tonic skin conductance	0.42 μ S	auto
Skin temperature	36.4 °C	auto
SpO ₂ offset	−0.5 %	guided
Skin → core offset	+0.6 °C	guided

STATUS FLOW

warming_up



active

with confidence %

Eight conditions, each with its own evidence

Alerts carry severity, the readings that fired, a plain-language explanation and suggested next steps.

Condition	Severity range	Signals that fire it	Tier
Fall Detected	critical	accel ≥ 2.8 g with tumble ≥ 2.4 rad/s plus post-impact stillness	Clinical floor
Low Oxygen	high $\rightarrow$ critical	SpO ₂ < 92 % / < 90 %, or accumulated hypoxic burden ≥ 3 %·min	Both tiers
Fever	medium $\rightarrow$ high	skin ≥ 37.8 °C or core-equivalent ≥ 37.8 °C, plus °C/h slope and HR coupling	Both tiers
Panic Attack	critical	stress index ≥ 0.72 with HR ≥ 30 bpm above activity expectation	Both tiers
High Stress	medium $\rightarrow$ high	stress index ≥ 0.60 or personal $z \geq 2.0\sigma$ on combined stress evidence	Both tiers
Tachycardia	medium $\rightarrow$ high	resting HR ≥ 110 bpm, severe from ≥ 135 bpm (absolute floor 140)	Clinical floor
Bradycardia	medium $\rightarrow$ high	HR ≤ 50 bpm at rest (absolute floor 38 bpm)	Clinical floor
Fatigue	low	HRV ≤ 20 ms or a ≥ 40 % drop versus the personal resting HRV	Both tiers
Inactivity	medium $\rightarrow$ high	no movement for the configured window, with a 24 h safety watch	Personal rule

ANATOMY OF ONE ALERT — WHAT A CAREGIVER ACTUALLY RECEIVES

CRITICAL

Fall detected — impact signature

02:15 · bathroom · GPS captured

“Acceleration spiked to 3.4 g with 2.9 rad/s of rotation, followed by 40 s without movement. Heart rate rose to 112 bpm afterwards. This matches a fall signature rather than a stumble.” $\rightarrow$ Recommended: call the wearer, then dispatch the primary contact.

COVERAGE OF THE LIVE SUITE

8 conditions + inactivity watch

2 independent trigger tiers

11 immutable clinical safety floors

Personalisation may add sensitivity, never remove it

Every condition is evaluated twice — against fixed clinical floors and against the wearer's own baselines.

TIER 1 — CLINICAL FLOORS (READ-ONLY)

Guideline constants that calibration can never move. A personal baseline must never make a clinical emergency look safe.

SpO₂ urgent / low 90 % / 92 % Core-equivalent fever ≥ 37.8 °C

Hypothermia ≤ 35.5 °C Urgent tachycardia ≥ 140 bpm

Urgent bradycardia ≤ 38 bpm Fall impact ≥ 2.8 g

Tumble rotation ≥ 2.4 rad/s

Immutable in the product: no caregiver edit, no calibration result and no model update can raise or lower these.

TIER 2 — PERSONAL σ -RULES (EDITABLE)

Deviations versus the wearer's own baselines — the tier that catches what a population threshold would miss.

Stress evidence $z \geq 2.0\sigma$

Fever deviation $z \geq 2.5\sigma$

Heart-rate deviation $z \geq 2.5\sigma$

HRV drop vs personal rest ≥ 40 %

Hypoxic burden ≥ 3 %·min

Inactivity window configurable

Live in Management → Calibration. Adjustable without a redeploy.

 Sensitivity slider shown to the caregiver: less sensitive ↔ more sensitive

THE TRIGGER RULE

**A condition fires when
EITHER tier fires.**

Personalisation is therefore an asymmetry in safety, not a trade-off: it can only ever add sensitivity.

Tier 1 fires

→ alert

Tier 2 fires

→ alert

Both fire

→ higher severity

Three layers of explanation, one language layer

Rules decide. Equations quantify. A language layer translates — and degrades gracefully.

01 Detection rules

DETERMINISTIC

Threshold and σ -gates decide whether an event exists at all. No probabilistic model sits between the sensor and the decision.

02 Equation evidence

QUANTIFIED

The alert stores its own evidence: z-scores, core-equivalent temperature, hypoxic burden, HR vs activity expectation.

03 Language layer

NARRATED

A hosted model turns that evidence into a caregiver-readable explanation and next steps — with a deterministic template fallback offline.

WHAT THE CAREGIVER READS

“Galvanic skin response rose to 8.4 μ S while heart-rate variability fell to 19 ms — the classic electrodermal signature of acute stress. Stress index is 0.74/1.00, well above Margaret's calm baseline.”

RECOMMENDED NEXT STEPS

- Call the wearer and confirm how she feels.
- Ask about pain, breathlessness or recent exertion.
- Escalate to the clinician if the index stays high for 20 minutes.

WHY A LANGUAGE LAYER — AND WHY IT MUST BE OPTIONAL

Adoption — Caregivers act on sentences, not on z-scores.

Auditability — Every generated line can be traced back to stored evidence.

Continuity — If the model provider is unavailable, detection still runs and alerts still write.

Cost control — Narration is the only call that costs money per event — and it is bounded by alerts, not by readings.

Verified, not asserted: the evidence in the repository

Three independent suites run against the live server — all green, all re-runnable on demand.

141

QA battery checks

51 UI · 1 session · 55 backend · 1
WebSocket · 8 sign-in · 25 end-to-end

PASS

74

Calibration-suite checks

Equations, calibration module, two-tier
OR-gates, HTTP API, demo triggers

PASS x2 · idempotent

17

DOM interaction checks

Sign-in regression, Calibration tab
panels, auto-fit, guided reference,
sliders

PASS

6

Bugs found and fixed

Each one reproduced, root-caused and
re-verified by a regression test

CLOSED

Defect found by the suites	Why it mattered	Resolution
Confirm dialogs always cancelled	Every delete / resolve confirmation silently did nothing	Decision flag added to the modal resolver
Buttons disabled after save	A save could leave the UI permanently locked	Element captured before the await
View / WebSocket lifecycle race	Slow renders wrote into dead DOM after navigation	Serialised navigation with null-guarded loaders
Demo stress scenario could not fire	The flagship demo alert never appeared	Scenario recalibrated; forced phases snap within one tick
Wrong password showed “session expired”	A simple credential error looked like a security problem	Login 401 now surfaces the real message
Demo drills swallowed by cooldown	Emergency drills silently suppressed for 10 minutes	Drill triggers clear the duplicate-suppression window

Reproduce in one command: `python3 calib_suite.py` against a running server — the suite is idempotent and safe to run repeatedly, which is what a diligence reviewer will ask for.

04

TARGET AUDIENCE

Who we build for

One wearer, one payer, one operator — each with a concrete job to be done.

Older adults at home

Family caregivers

Home-care and senior-living operators

Who we build for, in order

One wearer, one payer, one operator. Every segment is defined by a concrete job — not by a demographic.

PRIMARY

WEARER — NOT THE BUYER

Older adults aging at home

65+, living alone or with a partner, managing hypertension, COPD, diabetes or frailty.

- Wants to stay independent and not be “watched”
- Will not tolerate charging a device daily
- Needs the device to be invisible under a sleeve

Success metric: 20+ wear-hours per day, 90 % of days, after month one.

PRIMARY

BUYER + DAILY USER

Family caregivers

Adult children aged 40–60 and spouses who already act as the informal safety net from a distance.

- Already anxious between calls and visits
- Pays for peace of mind, not for a gadget
- Needs an answer in seconds during a scare

Success metric: alert acknowledged in under 5 minutes; subscription retained at 6 months.

SECONDARY

ENTERPRISE BUYER

Home care & senior living

Home-care agencies, assisted-living operators and care cooperatives running staffed caseloads.

- Needs multi-wearer oversight and an audit trail
- Buys workflows, not dashboards
- Exits trials that create extra staff work

Success metric: staff minutes saved per alert; pilot converts to paid renewal.

TERTIARY

FUTURE PARTNER

Remote care ecosystem

Remote-monitoring providers, payers and employers exploring supported self-management.

- Requires evidence of detection quality first
- Asks about data governance, retention and consent
- Moves slowly, contracts in multiples

Success metric: validated evidence pack and a signed integration pilot.

Three people decide whether this works

Adoption fails on the wearer's wrist, not in the demo. Design for all three at once.



Margaret, 78

The wearer

TECH COMFORT ●●●●

Hypertension and mild COPD · lives alone · two falls in 18 months

"I don't want to be a patient in my own house."

PAINS

- Wants to keep living at home without a nurse in the room
- Hates being reminded that she is old
- Will abandon anything that itches, buzzes or needs charging

WHAT EARNS THEIR TRUST

- Wrist-worn, unnoticeable, multi-day battery
- Alerts she can dismiss when she is fine
- No cameras, no constant calling



Sarah, 46

The family caregiver

TECH COMFORT ●●●●

Lives 90 minutes away · works full time · primary contact

"If something happens at 2 a.m., I need to know before the morning call."

PAINS

- Guilt about not being there
- Already drowning in group chats and check-in calls
- Cannot interpret a raw vitals chart

WHAT EARNS THEIR TRUST

- One screen with her mother's real state
- Plain-language explanation and a next step
- One-tap escalation with location



Dana, 39

The care manager

TECH COMFORT ●●●●

Runs 60 home-care clients · 14 field staff

"I need fewer surprises and an audit trail I can hand to a regulator."

PAINS

- Accountable for outcomes she cannot see
- Field staff resist extra data entry
- Must justify every subscription line

WHAT EARNS THEIR TRUST

- Caseload-level triage and trend reporting
- Evidence trail per incident
- Deployment that does not add work

What each person hires the product to do

Jobs-to-be-done mapped to the feature that answers it today — and to the metric that proves it worked.

Stakeholder	Job to be done	Feature serving it today	Proof metric
Wearer	"Let me keep living alone without feeling supervised."	Unobtrusive band, dismissible alerts, no cameras or microphones	Wear-days per month
Wearer	"Tell me when something is actually wrong."	Two-tier detection with wearer-facing status and vibration cues	Alert acknowledgement rate
Family caregiver	"Show me she is fine without me calling again."	Live Overview with vitals, motion state and 24 h trend sparklines	Sessions per week
Family caregiver	"When it goes wrong, tell me what to do."	Plain-language explanation, recommendation list, one-tap dispatch	Time to acknowledgement
Family caregiver	"Keep the record I will need later."	Incident timeline with GPS, dispatch log, downloadable history	Incidents with complete logs
Care manager	"Triage 60 clients without 60 phone calls."	Caseload-ready alert queue, severity sorting, per-wearer thresholds	Alerts resolved per staff hour
Care manager	"Defend the decision if something goes wrong."	Immutable clinical floors plus audit trail on every configuration change	Audit completeness
Clinician (reviewer)	"Understand why an alert fired before I act."	Evidence panel: z-scores, core-equivalent temperature, hypoxic burden	Clinician agreement rate

THE ADOPTION JOURNEY — WHERE MOST WEARABLES FAIL



05

ROADMAP

Roadmap & future features

What is shipped, what pilots next, and what the platform unlocks after that.

Shipped today — the working build

Next 0-12 months — pilot gates

Later 12-36 months — platform bets

From working prototype to validated product

Three horizons. Nothing moves to “next” until the gate before it is passed.

HORIZON 1

Shipped today

Working prototype

- Live telemetry over WebSocket
- 8 conditions + inactivity watch
- Two-tier detection with calibration
- Caregiver dashboard: 6 views
- Safety escalation with GPS + log
- Device pairing and MQTT test
- Roles, sessions, seeded demo data
- QA + calibration suites green

HORIZON 2

Next — 0-12 months

Pilot-hardening

- Field hardware revision with bench validation
- Push, SMS and voice delivery integrations
- Native mobile app for the wearer
- Caseload / multi-wearer operator view
- Alert precision telemetry and tuning loop
- Battery and wear-time analytics
- Consent, retention and audit controls
- FHIR-ready data export

HORIZON 3

Later — 12-36 months

Platform

- On-device equation engine at the edge
- Personal risk trajectory modelling
- Clinical review console for care teams
- Partner SDK and device-agnostic ingest
- Multi-condition chronic care programs
- Insurer and provider reporting packs
- Voice assistant and smart-home triggers
- Regulated pathway filing if evidence supports it

THE GATE BETWEEN HORIZONS

H1 → H2

Hardware accuracy validated on a bench and in 100+ wear-days

H2 → H3

Alert precision and retention targets met in pilot

H3 → scale

Economics and regulatory scope confirmed

Future features I — on the wrist

Wearer-facing additions. Each one is a hypothesis with the gate that would justify it.

On-device equation engine

Run stress index, fall physics and core-temperature estimation directly on the ESP32 for offline alerts when the network drops.

Gate: accuracy parity with the server path

Wear-time & battery intelligence

Track daily wear-hours and charge cycles in-product, with a low-battery escalation path that reaches the caregiver.

Gate: multi-day measured battery life

Sleep, recovery and circadian context

Turn overnight HRV, temperature and motion into a recovery picture that explains daytime alerts.

Gate: clinician review of the derived metrics

Multi-day cellular band

An LTE-M / eSIM variant for wearers with no home Wi-Fi — alerts travel even when the house does not.

Gate: unit cost and power budget

Wrist-worn form factor

Move from a band prototype to a validated, washable, skin-safe wearable enclosure.

Gate: DFM review and biocompatibility testing

Wearer-facing companion app

A deliberately simple wearer surface: status, dismiss, SOS, and nothing else.

Gate: wearer usability testing at 70+

Cardiac rhythm screening

Extend PPG analysis towards irregular-rhythm flags and AF burden estimates.

Gate: reference ECG comparison study

Medication and routine adherence

Gentle nudges tied to the wearer's routine, correlated with physiological response afterwards.

Gate: adherence improvement measured in pilot

Future features II — on the caregiver's screen

Everything that shortens the distance between an alert and a human response.

Push, SMS and voice escalation

Gate: partner integration + delivery testing

Delivery receipts, retry ladders and a defined fallback order, so an urgent alert has more than one way to arrive.

Care-team messaging

Gate: pilot feedback on workflow value

Discuss an alert in context, attach the evidence and keep the thread attached to the incident record.

Caseload console for operators

Gate: 2-3 signed pilot operators

One screen for a care manager's entire roster: triage queue, severity sorting, per-wearer drill-down and shift handover notes.

Scheduling and visit context

Gate: rostering integration with a pilot partner

Link readings to scheduled visits and to the staff member on duty, so an alert lands with the right person.

Family circle and delegation

Gate: consent model + legal review

Time-boxed access for a sitter or a visiting nurse, with an audit trail showing who saw what.

Multilingual, accessible surfaces

Gate: translation review per market

Localised explanations and a large-text mode for older caregivers — the same alert understood by everyone in the circle.

Weekly narrative reports

Gate: clinician feedback on usefulness

A written summary per wearer: what changed, how it trended, what was actioned — readable by a clinician.

Explainable alert audit

Gate: quality-review workflow in pilot

A reviewer view that shows why any historical alert fired, with its stored evidence and the thresholds in force at that moment.

Delivery integrations are a partner dependency, not just a build task — escalation providers, local emergency numbering rules and consent policy differ per market and must be validated before any launch claim.

Future features III — platform, data and evidence

The bets that turn one product into an ecosystem — and the evidence that must come first.

Capability	What it adds	Why it matters commercially	Horizon
Partner SDK & device-agnostic ingest	Any sensor payload can be normalised into the same pipeline	Turns competitors' hardware into distribution	Later
Clinical review console	Clinician view of trends, evidence and alert history per wearer	Unlocks provider channel and referral loops	Later
Personal risk trajectory	Longitudinal baselines modelled into drift and deterioration signals	Shifts from alerting to prediction — the defensible layer	Later
FHIR / HL7 export	Structured, standards-based sharing with care systems	Removes the largest integration objection from operators	Next
Reimbursement evidence pack	Outcome reporting designed for payer questions	Converts a subscription into a funded care cost	Later
Data governance programme	Retention controls, consent ledger, least-privilege access, threat model	Precondition for any health-system conversation	Next
Federated personalisation	Improve equation priors across consenting wearers without moving raw data	Better day-one accuracy for every new wearer	Later
Multi-condition programmes	COPD, cardiac and post-surgical monitoring tracks on one platform	Expands account value without new hardware	Later

STANDING RULE FOR EVERY FUTURE FEATURE

Evidence first — No claim ships before the measurement that supports it.

Human in the loop — Every escalation ends with a person, never with an automated decision.

Safety floors stay — No roadmap item can move a clinical constant or a consent boundary.

06

BUSINESS & ECONOMICS

Business model

Hardware opens the account; subscriptions and operator contracts fund the company.

Device sale at \$149 ASP

Subscription at \$12 / month

Operator licence per wearer

Hardware opens the account. Software keeps it.

Device margin funds acquisition; subscription and operator contracts fund the company.

DEVICE SALE

\$149

one-time

Band, charging cradle and starter kit.
Priced at roughly 1.7x unit cost —
deliberately not the profit centre.

CONSUMER SUBSCRIPTION

\$12 / month

recurring

AI insights, unlimited history,
caregiver circle and safety escalation.
The primary revenue engine.

OPERATOR LICENCE

**\$9-18 / wearer /
mo**
contract

Caseload console, staff workflow,
reporting and audit trail for
home-care and living operators.

EVIDENCE & INTEGRATION

Pilot fee + licence

strategic

Paid pilots with providers and payers
that convert to integration contracts
once detection quality is proven.

WHY A SUBSCRIPTION IS DEFENSIBLE HERE

- Value is continuous, not transactional. The safety net only exists while the service runs.
- Switching cost compounds. Calibration baselines and history belong to this wearer.
- Marginal cost is low. Storage and narration per wearer are bounded by alerts, not readings.
- Operators need the software. The hardware alone cannot serve a caseload.
- Price sits below the alternative. A private caregiver hour costs far more than a month.

PRICING PRINCIPLES WE WILL TEST IN PILOT

Annual prepay — discount to reduce churn and fund hardware

Family circle — up to 3 contacts included, more as an add-on

Operator tiering — per-wearer price steps down with volume

Hardware bundling — band free on a 24-month subscription?

No charge for alerts — monetising safety events would be indefensible

Unit economics: where the money is made and lost

Illustrative base-case assumptions for discussion — not validated costs.

DEVICE UNIT — BASE CASE

Line	Amount	Note
Average selling price	\$149	consumer, one-time
Bill of materials	−\$62	band, sensors, ESP32, enclosure
Assembly and test	−\$12	contract manufacturer
Packaging and freight	−\$8	consumer retail pack
Warranty and returns reserve	−\$8	3 % return assumption
Gross contribution per device	\$59	39 % gross margin

Target: \$52 device COGS at 10k units, improving with procurement scale.

SUBSCRIPTION UNIT — PER WEARER, PER MONTH

Line	Amount	Note
List price	\$12.00	consumer tier
Cloud, storage and narration	−\$0.85	bounded by alerts, not readings
Support and service	−\$1.60	the real cost driver
Payment, tax, bad debt	−\$0.55	≈ 4.5 % of revenue
Contribution per wearer / month	\$9.00	75 % gross margin
Blended CAC target	\$85–140	payback in 10–16 months

LTV : CAC ≈ 3.2x at a 24-month average life, before the device contribution is counted.

THE FOUR NUMBERS THAT DECIDE THIS BUSINESS

Alert precision

> 90 %

alerts that a caregiver judges actionable

Time to acknowledgement

< 5 min

median, from alert to human response

Wear adherence

> 20 h/day

sustained after the first month

Support cost per wearer

< \$1.60

the margin line most likely to break

Prove the loop in a narrow pilot, then scale

Success means a trusted response loop, repeat use, and a cost to serve that survives scale.

0-3 MONTHS

PILOT DESIGN

- 2-3 pilot partners signed
- 100-150 target wearers
- Caregiver + wearer interviews
- Safety review of escalation rules

3-9 MONTHS

CONTROLLED PILOT

- Pre-registered endpoints
- Adherence and alert telemetry
- Weekly service feedback loop
- Shadow-mode comparison of thresholds

9-18 MONTHS

REPEATABLE SALES

- Convert paid sites
- Family-plan pricing tests
- Channel and support playbook
- Second vertical preparation

CHANNELS, IN ORDER OF INTENT

- 1 Direct pilot conversions** — The first revenue comes from sites that already tested the product.
- 2 Family caregiver communities** — Support groups, condition associations and referral from existing families.
- 3 Home-care operators** — Caseload deployments where one contract places many wearers.
- 4 Clinician referral** — Discharge and chronic-care follow-up, once the evidence pack exists.
- 5 Pharmacy and retail** — Late-stage consumer reach; expensive before proof.

PILOT SCORECARD — WHAT WE MEASURE

Wear-days / active days
adherence in real life

Battery days per charge
the form-factor gate

Alert precision (PPV)
did the alert deserve action?

False alerts per day
the churn driver

Time to acknowledgement
is the loop actually closed?

Contact delivery success
does the escalation path work?

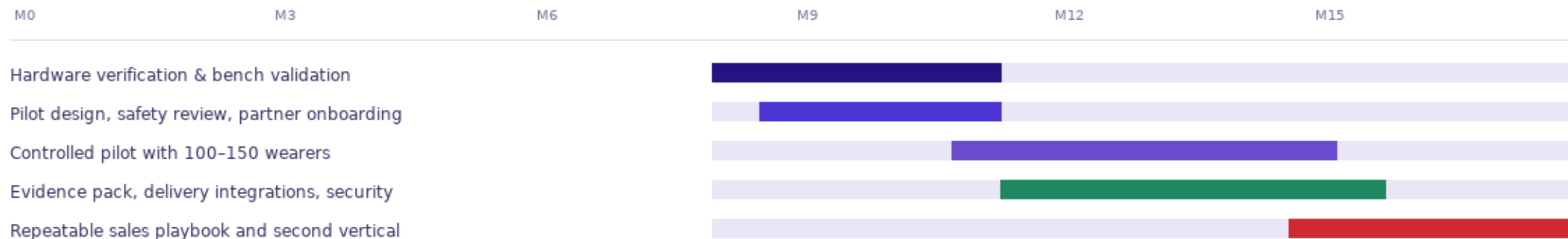
Willingness to pay
price sensitivity, tested directly

No signed partner is represented here — the pilot funnel is the immediate workstream.

The pilot is the product: an instrument, not a demo

Designed to surface failure modes — a pilot that only proves the happy path wastes the money.

18-MONTH RUNWAY PLAN — WHAT HAPPENS WHEN



PILOT DESIGN — THE NON-NEGOTIABLES

- **Pre-register the endpoints. Precision, sensitivity, time-to-response — defined before data.**
- **Shadow mode first. Run detection silently against reference measurements before alerting families.**
- **Human review of every alert. A clinician or nurse adjudicates each event in the first weeks.**
- **Instrument the awkward cases. Charging, showering, sleeping, travel — adherence lives there.**
- **Consent and retention from day one. The pilot is where the privacy model must prove itself.**

FAILURE MODES THE DESIGN MUST CATCH

Risk	Design response
Measurement bias / sensor noise	Bench testing against reference instruments; per-signal quality checks
False alerts erode trust	Shadow mode, adjudication, tuning loop with weekly review
Battery or wear abandonment	Daily wear-time telemetry; usability interviews; charger redesign
Emergency delivery failure	Delivery receipts, retries, fallback contacts, honest disclaimers
Privacy or security incident	Least privilege, retention limits, threat model before pilot start
Regulatory scope drift	Define intended use as wellness / safety support, not diagnosis

07

FINANCIALS & FINANCING

Financial plan & the ask

An illustrative five-year plan and a \$1.5M pre-seed to buy evidence, not features.

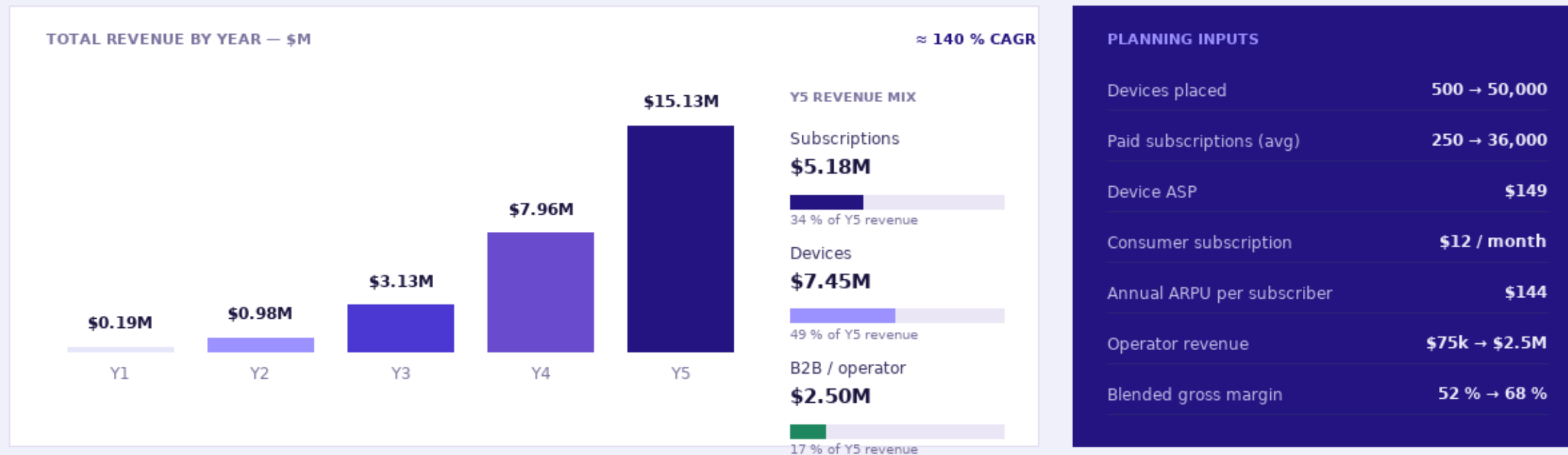
Break-even planned in year four

\$1.5M for ~18 months

Milestone: validated, pilot-ready

Five-year planning model: revenue build

Illustrative base case. Assumptions are planning inputs, not forecasts, guidance or commitments.



WHAT HAS TO BE TRUE — AND WHAT IS DELIBERATELY NOT MODELLED

- Hardware cost reaches \$52 per unit at 10k volume and yield holds.
- Support cost stays under \$1.60 per wearer per month as the base grows.
- Subscription churn below 2 % per month after the first 90 days.

Not modelled: reimbursement, acquisition of a competitor, hardware margin expansion beyond scale, or any enterprise contract above \$300k.

Excluded: taxes, financing costs, depreciation, working capital, warranty tail and R&D capitalisation.

Profit and loss: break-even in year four

Planning scenario only. Break-even timing is highly sensitive to adoption, support cost and churn.

\$M, except margin and headcount	Y1	Y2	Y3	Y4	Y5
Device revenue	0.07	0.45	1.49	3.87	7.45
Subscription revenue	0.04	0.23	0.92	2.53	5.18
Operator / pilot revenue	0.07	0.30	0.72	1.55	2.50
Total revenue	0.19	0.98	3.13	7.96	15.13
Cost of revenue	-0.09	-0.44	-1.26	-2.73	-4.69
Gross profit	0.10	0.53	1.87	5.23	10.45
Gross margin	52 %	55 %	60 %	66 %	69 %
Operating expense	-0.70	-1.25	-2.20	-3.60	-5.80
EBITDA (planning)	-0.60	-0.72	-0.33	1.63	4.65

GROSS MARGIN TRAJECTORY — WHY IT IMPROVES



THE THREE CONDITIONS FOR BREAK-EVEN

Mix shifts to software — subscription share rises from 24 % to 34 % of revenue

Hardware stops losing money — COGS falls from \$90 to \$52 per unit

Support scales sub-linearly — automation before headcount

The ask: \$1.5M to buy evidence, not features

An 18-month plan to reach a validated, pilot-ready product with measurable endpoints.

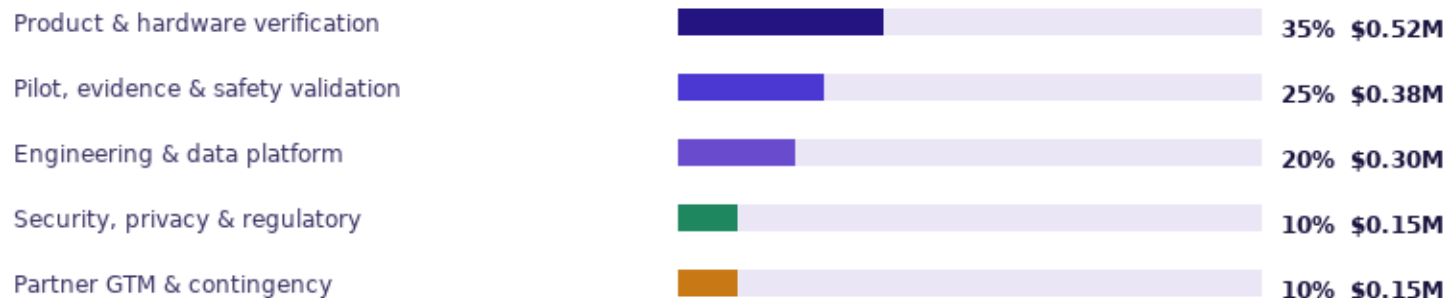
RAISE

\$1.5M

pre-seed

≈ 18 months of runway to a validated, pilot-ready milestone.

USE OF FUNDS — PLANNING ALLOCATION



WHAT THE MONEY BUYS — MILESTONES AT MONTH 18

2-3

pilot partners signed with written protocols

100-150

target wearers generating continuous data

< 5 min

median time from alert to acknowledgement

> 90 %

target alert precision, adjudicated by clinicians

1

evidence pack suitable for a provider or payer conversation

USE OF FUNDS IS A HYPOTHESIS TOO

Allocation shown is a planning view. Raise size, valuation, instrument and timing require founder and legal confirmation, and will be revisited after the pilot readout — capital should follow evidence, not precede it.

Risks named early, with the response already designed

A pilot exists to test failure modes. These are the six that would end the company if unmanaged.

Measurement bias and noise



Prototype sensors drift with skin, motion and temperature.

Response: Bench validation against reference instruments; per-signal quality gates; subgroup review.

False alerts erode trust



One bad night-time alert can end a family's willingness to continue.

Response: Shadow mode before alerting; clinician adjudication; tuning loop with a published false-alert target.

Battery and wear abandonment



A band that is not worn cannot save anyone.

Response: Daily wear-time telemetry; usability interviews; charger and enclosure redesign in Horizon 2.

Emergency delivery failure



A missed escalation is the most serious failure mode in the product.

Response: Delivery receipts, retry ladder, fallback contacts, explicit non-emergency disclaimers, human confirmation.

Privacy and health-data exposure



Health data attracts both regulation and attackers.

Response: Least privilege, credential rotation, retention limits, threat model and audit logging before pilot start.

Regulatory scope drift



A wellness product can drift into a medical device claim by accident.

Response: Define intended use; no diagnostic language; clinical and legal review before each pilot.

PRODUCT GUARDRAIL — NON-NEGOTIABLE

NeuroLink Wear supports wellness and safety decisions made by humans. It is not a diagnostic device, does not replace emergency services, and never presents a single sensor reading as a clinical conclusion.

NEXT STEP

Make the signal useful to someone.

Wearable context. Explainable alerts. A safer path to human response — and the evidence to prove it before we ask anyone to trust it.

PILOT PARTNERS

Home-care or senior-living operators willing to run 100–150 wearers with written endpoints.

CLINICAL VALIDATION SUPPORT

Advisors to pre-register endpoints and adjudicate alerts during the pilot.

INVESTORS

\$1.5M pre-seed aligned with a pilot-led, evidence-first plan.

THE PLAN IN ONE COLUMN



Today

Working prototype: live telemetry, two-tier detection, caregiver workflow, green test suites.



Next 6 months

Hardware verification, delivery integrations, pilot onboarding with 2–3 partners.



Next 18 months

100–150 wearers instrumented, precision and response-time evidence, paid renewals.



Then

Operator contracts, remote-care partnerships, platform expansion on proven detection quality.

CONTACT — EDIT BEFORE SENDING

founders@neurolinkwear.example · +00 000 000 000

Replace with the real contact details.